LESSON 5.1

RANDOM SAMPLES VS. POPULATION

LESSON INTRODUCTION

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.

LEARNING TARGETS

- I can determine if a sample is representative of the population.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.DP.1.4 Estimate a population total, mean, or percentage using data from a sample survey and develop a margin of error through the use of simulation.

ESSENTIAL QUESTIONS

- How can you collect data about a large population?
- How do you choose a sample?
- How do you ensure a sample is representative of the population?

LESSON NARRATIVE

In this lesson, students are introduced to surveying a population using a representative sample. They are guided through understanding the difference between the population and a sample and determining the best means for ensuring that they are collecting data from a representative sample. Students then collaborate and explore to further develop the idea of representative samples, including exploring how the size of the sample can impact its accuracy reflecting the population.

COMPONENT SUMMARY

5.1.1 WARM-UP *(~5 MIN.)*

Career Profile – electric vehicle technician

5.1.2 GUIDED INSTRUCTION *(10 MIN.)*

Introduction to choosing representative samples of a population

5.1.3 COLLABORATION *(10-15 MIN.)*

Collaborative exercise in comparing sampling methods and suggesting better methods

5.1.4 EXPLORATION *(10 MIN.)*

Explore how sample size affects results

5.1.5 YOUR TURN! *(5 MIN.)*

Choose, describe, or critique sampling methods used for given populations of interest

5.1.6 WRAP-UP *(~5 MIN.)*

Student assessment of choosing a representative sample

RESOURCES

GLOSSARY

Key Terms:

- Population
- Sample
- Representative
- Random sampling

Additional Terms:

- N/A

MATERIALS

- Career Profile video
- (Optional) Calculator
- (Optional) Premade circles for 5.1.4 Exploration
- (Optional) Markers

ADDITIONAL PREPARATION

- 5.1.1 Warm-Up contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse population with sample.
- Students may struggle with the idea of a sample being representative of a population.
- Students may not be able to predict how different methods of sampling could lead to different and misleading conclusions.

THINGS TO CONSIDER

- Once students have worked on the first problem polling voters in FL, consider stopping and doing a check with them to make sure they are on the right track.
- Consider creating a visual representation of the new vocabulary to use as an anchor chart or on a word wall.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

5.1.3 Collaboration gives students the perfect opportunity to apply these new concepts to different contexts. Consider applying the Think-Pair-Share routine for both scenarios to create opportunities for students to solidify their understanding of the difference between a sample and a representative sample.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

Students are being introduced to the idea of samples of a population as well as representative samples. Because this lesson is based around the vocabulary, guide students to discuss and reflect on these ideas in order to more fully develop them. Both the 5.1.3 Collaboration and 5.1.4 Exploration provide opportunities for students to discuss and reflect on their ideas about these concepts in context.

DIFFERENTIATE INSTRUCTION

5.1.4 Exploration implements both auditory and visual methods into the activity. Consider providing drawing implements for students to complete their spinners to aid in their understanding.

Additional differentiation opportunities are denoted throughout the lesson guide.

5.1.1 WARM-UP: ELECTRIC VEHICLE TECHNICIAN

QUESTIONING

Do you know someone who is an electrical engineer or an electric vehicle technician? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider allowing students to research fields where electrical engineers can work, choosing one of interest and creating a summary based on their learning either for homework or extra credit.

5.1.1 Warm-Up: Electric Vehicle Technician

Ben Grandy is an electrical vehicle technician for City Car Share. When Ben was considering what to do with his life, he knew he wanted to do something that would bring him happiness. Now he uses the skills he has gained from his study of landscape architecture and work as an electrical engineer to maintain the fleet of electric vehicles and electric bikes used by City Car Share.

Ben found that dedicating himself to a cause that is bigger than himself brings him happiness.

1. Think about what you would like to do with your life. What kind of work would make you happy?

   Answers will vary. I would like to be a teacher because I like to help people. I would like to be a doctor because I'm curious about the brain.

In talking about his work, Ben said, "That's how engineering and design is. You never get to a point where you think, 'Oh, that's perfect! That's done.' You start asking questions like, 'Now that we've gotten this far, what's the next step? How can we improve this?'"

2. How do you think having this mentality is helpful?

   Answers will vary. This mentality is helpful because it means you can always get better. It is helpful because it means that you will never be lazy or complacent.

5.1.2 Guided Instruction: Is the Sample Representative of the Population?

Many organizations want to know how large groups of people think and behave. This helps businesses make decisions regarding what to sell, when to be open, and whom to target with advertisements. Research is also used to make predictions about events like elections, and to help doctors provide appropriate healthcare to their patients.

To gather this information, companies and organizations often collect data ***samples*** from a ***population*** they are studying.

A ***population*** is the entire set of cases or individuals under consideration in a statistical analysis.

A ***sample*** is a subset, or part, of a population.

1. Imagine a group of researchers want to gather information about which candidate for sheriff all registered voters in Glades County, Florida plan to vote for in an upcoming election.

   Use the definitions to complete the statements.

   In this scenario, the [○ population / ○ sample] being studied is all registered voters in Glades County, Florida. It is not realistic for researchers to ask every single registered voter, so researchers would collect data from a [○ population / ○ sample] of registered voters in Glades County, Florida to make predictions.

2. Volusia County School District wants to find out more about which mode of transportation students take to school, so they decide to conduct a survey. The entire school district has a population of 63,223 students. Since the number of students is so large, the district will only survey some of the students.

   a. In this scenario, what is the population being studied?
      Students in the Volusia County School District

   b. What is the sample being used?
      The students that are interviewed for the study.

To ensure the students surveyed are representative of all students in Volusia County, a system is used to select who is surveyed.

A sample that is **representative** of a population resembles the population and can be used to make accurate predictions about the population.

3. Twenty-five students from each grade level will be chosen at random to participate in the survey. With your partner, discuss how you think this method will help ensure the sample is representative of the population.

   Answers will vary. This will help ensure that the sample is not biased, however that size is a bit small to ensure that there is true variability.

5.1.2 GUIDED INSTRUCTION: IS THE SAMPLE REPRESENTATIVE OF THE POPULATION?

TEACHER MOVES

This is the first time that students are introduced to the terms population, sample, representative sample, and random sampling. While modeling the problems in this component, include developing a strong understanding of these terms.

ENRICHMENT

- What is the group the researchers are studying?
- Does this group represent the population or the sample?
- Whom are the data from?
- Does this group of people represent the population or the sample?

TEACHER MOVES

Consider having students connect the idea of a representative sample to the idea of representatives in real life. The whole school doesn't meet to make decisions; instead, a representative of each grade is elected to a council to speak for that grade. Each state can't send its entire population to the house of representatives to vote, but the representative goes and votes for them.

ENRICHMENT

A representative sample is one where all parts of the population are represented.

- Would it be a good representation if only the boys were surveyed?
- What about only the people with fall birthdays?
- How can we determine what a good representative sample might be?

5.1.2 GUIDED INSTRUCTION: IS THE SAMPLE REPRESENTATIVE OF THE POPULATION? (CONTINUED)

ENRICHMENT

If students struggle to explain why the sample is not representative, guide their thinking through questioning:

- If Claire is asking about education levels on a college campus, who will she most likely be talking to?
- If someone already has a degree, why would they be in the campus library?
- If someone chose a line of work that does not require a degree, why would they be in a campus library?

QUESTIONING

Consider challenging students to discuss how Claire could find a representative sample to survey.

4. Claire wants to collect data on the highest eduation level achieved by adults in Jacksonville, Florida. She decides to go to the University of North Florida campus and ask the first person who comes into the library every **30** minutes about their education level.

   a. What is the population?

   The population is adults in Jacksonville, Florida.

   b. What is the sample?

   The sample is the first person coming into the UNF library every thirty minutes.

   c. Explain why a sample chosen in this way may **not** be representative of the population.

   The sample will most likely be college-aged students, rather than adults of all ages. There also may be a disproportionate representation of college-educated adults since it is taking place on a college campus.

A ***random sampling*** is a smaller group of people or objects chosen from a larger group or population by a process giving equal chance of selection to all possible people or objects, and all possible subsets of the same size.

5.1.3 Collaboration: Comparing Methods for Selecting Samples

For each scenario below, read the information about the study and answer the questions that follow.

Scenario 1: Lin is running for president of the seventh grade. She wants to predict her chances of winning the election. She is considering three different methods for surveying a sample of the students.

1. Complete the table to describe the benefits and problems with each of the possible survey methods described.

Sample Surveyed	Benefits of this Method	Problems with this Method
Method 1: Ask everyone on her basketball team who they are voting for.	*This method is convenient.*	*Her teammates may not respond to her honestly. There may be students who aren't in seventh grade on her team.*
Method 2: Ask every third girl waiting in the lunch line who they are voting for.	*This will provide a good sample size.*	*The sample is not representative of the population – only female, and they may not be seventh graders depending upon the lunch period.*
Method 3: Ask the first 15 seventh graders to arrive at school one morning who they are voting for.	*This method includes the voting demographic (seventh graders).*	*Students who arrive early at school may be part of the same subgroup (arriving early for a club meeting or practice). It is also a small sample size.*

2. How could the different methods for selecting a sample in Scenario 1 lead to different conclusions about the population?
Method one would highlight how basketball players feel about the election – they likely would vote for Lin because they are teammates, so this would be a biased sample. Also, not all of these students would be able to vote for her as they may be in different grades.
Method two would provide a sample of girls of different grades at lunch time. It does not include boys, who may vote differently, and it may include some non-seventh graders.
Method three selects the target grade group, but might only include students who are dropped off / walk to school. It is also a very small sample, so the feelings of the general seventh grade population may be very different.

3. Explain which of the methods listed in the table would be the most likely to produce a sample that is representative of the population.
The third method is the most targeted – it specifies seventh graders. It could be made stronger if Lin chose students at random throughout the morning – students arriving at the same time are likely to be neighbors or friends.

4. With your partner, determine a better way to select a sample for this study.
Student responses will vary – a better way to select a sample would be to target random groups of seventh graders – perhaps to survey every fifth seventh grader that enters in the morning.

5.1.3 COLLABORATION: COMPARING METHODS FOR SELECTING SAMPLES

TEACHER MOVES

Think-Pair-Share

To encourage student discussion, consider having students read each scenario first and use a Think-Pair-Share to discuss the answers. Students should be collaborating throughout the activity. Monitor to identify any misconceptions, especially in regard to representative samples, to identify any students who might need further clarification.

ENRICHMENT

For students who may still be struggling with the idea of population versus representative sample, consider asking:

- Who or what is the population?
- Who or what would the sample be?
- Who or what does the sample NOT include?

By asking who or what is left out of the sample, students can think of "representative" in another way, for example, as in not leaving specific groups of the population out.

5.1.3 COLLABORATION: COMPARING METHODS FOR SELECTING SAMPLES (CONTINUED)

QUESTIONING

For students who finish early, consider directing them to find another way to choose a representative sample for scenario 2.

Scenario 2: Maneet, a nutritionist, wants to collect data on how much caffeine the average American drinks per day. She works with three of her colleagues to decide how to survey a sample of the population.

Method 1	Method 2
Maneet and Diarmaid believe the best way to survey is to ask the first 20 adults who arrive at a grocery store after $10:00$ A.M. about the average amount of caffeine they consume each day.	Faith and Santiago would rather ask the first adult who comes into a coffee shop every 30 minutes about the average amount of caffeine they consume each day.

5. What recommendation(s) would you give to this committee about how to select the most representative sample?

   In the first method, you are likely to have a sample that has similar profiles (early morning coffee drinkers.) While it's better to conduct the survey at a neutral location like a grocery store, it would be better to conduct it throughout the day and with more random selections (i.e. every __th adult who comes in).

   With the second method, there is likely going to be some bias because the survey is being conducted at a coffee shop. Adults going to a coffee shop may be likely to consume more caffeine. They should change their location but keep the randomness of their selection method.

5.1.4 Exploration: Sample Size

Isabella is playing a computer game. Her challenge is to create a replica of a hidden spinner that she cannot see. Her score is based on two things: the accuracy of the spinner she creates and the number of spins she uses to create the spinner (the fewer spins she uses equals a higher score).

Isabella creates her first replica after 12 spins of the hidden spinner. Below is a table showing the outcomes of the first 12 spins.

Color	Number of Spins
red	2
purple	4
blue	3
yellow	3

1. Use the data from the first 12 spins to draw a sketch of what you think the hidden spinner might look like.

5.1.4 EXPLORATION: SAMPLE SIZE

TEACHER MOVES

The objective is not to have a perfect circle but rather a model that represents the data in the table. It may be helpful to have students identify the population (the spinner) and the sample (the 12 spins) before completing the problem in order to connect the lesson's concepts.

DIFFERENTIATE INSTRUCTION

As stated above, the objective isn't about the circle so much as the model. Therefore, it may be helpful for students that are struggling to be provided with an already created circle that is split into 12 sections in order for students to focus more on the idea of the sample versus interpreting data.

5.1.4 EXPLORATION: SAMPLE SIZE (CONTINUED)

DIFFERENTIATE INSTRUCTION

Students don't need to worry about perfectly splitting the circle into 36 sections. If students are struggling, facilitate thinking of the number of spins as part of a whole. Therefore, the green spins were 9 out of 36 which is equivalent to a fourth of the circle. Choose the data that is easiest to visualize to section the circle first, then fill in with the other sections. Again, consider providing a circle to students if desired.

TEACHER MOVES

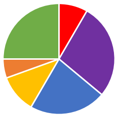
Facilitate student discussion toward the understanding that too small of a sample can lead to part of the population not being represented at all.

- How does this table compare with the last?
- Which colors weren't even represented in the first table?
- Looking at the data, which color would you assume takes up the most space on the spinner?
- And the least space?
- How does that compare with your first spinner?

Isabella's spinner was not accurate, so she had the computer generate additional data. The table below shows the data from the first 36 spins of the spinner, including the 12 shown in the previous table.

Color	Number of Spins
red	3
purple	10
blue	8
yellow	4
orange	2
green	9

2. Revise your spinner based on this new data.



3. Explain which spinner you think is closest to what the hidden spinner looks like.

   The second spinner seems more accurate. There were more colors than we had previously known, and the distribution was more varied. There was also more data available to construct the second spinner. A larger sample gives more information.

4. The second spinner Isabella drew was far more accurate than the first. Having additional spins gave a more accurate representation of what the spinner actually looked like. What can this simulation teach us about choosing the size of the sample in a population study?

   This shows us that it's important to have a large sample – otherwise we can draw inaccurate conclusions.

5.1.5 Your Turn!

1. The meat department manager at a grocery store is worried some of the packages of ground beef labeled as having one pound of meat may be under-filled. He decides to take a sample of 5 packages from a shipment containing 100 packages of ground beef. The packages were numbered as they were put in the box, so each one has a different number between 1 and 100.

   Describe how the manager can select a fair sample of 5 packages.

   The manager could choose multiples of 20. Then he would have five samples that had been spaced out throughout the order.

2. Diego wants to survey a sample of students at his school to learn about the percentage of students who are satisfied with the food in the cafeteria. He decides to go to the cafeteria on a Monday and ask the first 25 students who purchase a lunch at the cafeteria if they are satisfied with the food.
   Explain whether you think this is a good way for Diego to select his sample.
   Diego should consider choosing students at random rather than the first twenty-five students. The students who arrive early to lunch are most likely the ones who are excited about lunch and would skew his data positively.

3. Eleanor wants to make the most accurate prediction about who will win the student council election. Explain whether it would be better for her to conduct a random sample of 10 people or 50 people.
   A sample of fifty people would provide a bigger and thus more accurate predictive sample.

5.1.5 YOUR TURN!

DIFFERENTIATE INSTRUCTION

For students who might struggle verbalizing the best way to choose a representative sample, provide sentence stems or targeted questions for each scenario.

How could he ensure that he is checking packages randomly throughout the order?

The meat department manager could choose to sample every _________ package. What kind of opinions might the first 25 people in line have? If Diego chooses the first 25 students in line, ___________________________. How does the size of a sample impact how well it represents the population? Eleanor should choose ________ because it is likely to better represent the population as it is a __________ sample.

5.1.6 Wrap-Up: Cool Down

Anthony is conducting research to determine how students at his school get to school each day (biking, walking, car, bus, etc.). Because there are many students in his school, he will be conducting a survey of a random sample of students to make predictions about the overall population.

1. Select all the options that would produce a random sample to represent the overall population of the school.

   - ☐ Anthony will get to school before everyone else and survey the first 50 people who walk in the doors.
   - ☐ Anthony will survey everyone at his jazz band practice.
   - ☐ Anthony will survey two randomly chosen students in every homeroom.
   - ☐ Anthony will assign each student in the school a number, randomly choose 50 numbers, and survey those students.
   - ☐ Anthony will survey every third student he sees on his walk to school.

5.1.6 WRAP-UP: COOL DOWN

QUESTIONING

For students who may finish early, challenge them to describe how the other options could result in misleading survey results.